STATISTICAL ANALYSIS OF DOG REHOMING TIMES ACROSS THREE BREEDS

Coursework Report – MATH5741M: Statistical Theory and Methods

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1. Introduction

The main objective of this report is to analyze if the true mean rehoming time in shelters coincides with the theoretically proposed figure of 27 weeks and if this rehoming time is affected by breeds. This belief was investigated with the usage of a randomly created sample of 3 dog types (Mixed Breed, Shih Tzu, and Staffordshire Bull Terrier). Despite the main interested variable is the number of weeks a dog spent in the shelter before being rehomed, the data also represents other features such as health scores of each dog, age category, and return status by providing some handy information regarding the dogs in the sample.

Generally, the analyses on the following questions have been performed:

- What is the typical time for each breed to be rehomed?
- What are the best probability distributions for rehoming times?
- Is mean rehoming period equal to the 27-week per breed?

Investigating these points can help us to define if breed has an effect in adaptation period, and discussing some limitations and insights may be helpful for shelters in their future planning.

2.Results

2.1 Data Cleaning Summary

The original dataset contained a total of 956 observations. Looking carefully, missing values for rehoming times were removed along with some entries containing negative figures for *Rehomed* or *Visited columns*, which were regarded as impossible in this case. Missing breed labels were also excluded as dogs must be assigned to one of these breeds. Finally, some duplicated entries were removed, which might bias any predictions and comparisons.

In total, 32 rows (3.3%) were removed: 10 duplicates, 9 with missing rehoming time, 6 with missing breed, and 7 with negative time values. Hence, this left 924 whole observations for analysis.

2.2 Numerical and Graphical Summaries

Observations were separated into 3 groups which are Mixed Breed (n = 694), Shih Tzu (n = 24), and Staffordshire Bull Terrier (n = 205). The following table (**Table 1)** shows the key summaries of these breeds. It is clear that, rehoming times were quite similar across groups. The interquartile ranges overlapped strongly, and all breeds

showed mild positive skewness which suggests that most dogs were adopted within a typical timeframe whereas a small number experienced relatively longer stays.

Table 1. Summary of key characteristics by breed

Measure	Mixed Breed (n= 694)	Shih Tzu (n =24)	Staffordshire Bull Terrier (n = 205)
Rehomed (weeks)	Mean: 19.11 SD: 9.49 Median: 18 IQR: 12–25	Mean: 19.50 SD: 8.83 Median: 17 IQR: 12–25	Mean: 19.46 SD: 10.53 Median: 18 IQR: 11–26
Visited (weeks)	Mean: 14.59	Mean: 16.67	Mean: 13.13
Health condition group	Poor: 15.8% Moderate:76.4% Good: 7.8%	Poor: 16.7% Moderate:83.3% Good: 0%	Poor: 15.6% Moderate: 69.8% Good: 14.6%
Age distribution	Fully grown:98.7% Puppy: 1.3%	Fully grown:58.3% Puppy: 41.7%	Fully grown: 94.6% Puppy: 5.4%
Average rehoming time for returned vs non-returned dogs	No: 19.3 weeks Yes: 17.8 weeks Unknown: 20.2 weeks		

These patterns are confirmed again in **Figure 1**, where the boxplots for all three breeds have almost identical medians and IQRs. Mixed Breed and Staffordshire Bull Terrier dogs represent a few higher outliers whereas the overall distributions are greatly similar. Even though the Shih Tzu type is a little less variable, this is mainly because of the limited observations rather than an actual difference in pattern.

In terms of the age category, most dogs were fully grown before being rehomed. In addition, health scores categorized into Poor, Moderate and Good health conditions, where most dogs rehomed were in moderate health condition. Return status of the dog also showed only minor differences and these figures can be seen in **Figures 2-4**.

Overall, the numerical and graphical summary show that the three breeds are generally comparable, and there is currently no strong gap between rehoming values.

2.3 Modelling and Estimation

Rehoming numbers are all positive and mildly right-skewed for all three breeds. Hence, the Gamma and Lognormal distributions were regarded as logical models. Histograms with fitted curves (**Figure 5**) were plotted and they illustrated that both Gamma and Lognormal models captured the overall pattern of the 1st and 3rd breeds considerably well although the 2nd histogram appeared slightly bimodal, though this is probably the result of the small sample size (n = 24).

Normal distribution was rejected here as its symmetry make it irrelevant for positively skewed data. Maximum likelihood estimates were acquired for both distributions based on each breed, which served as the foundation for further diagnostic analysis.

2.4 Model Checking

Model suitability was tested utilizing Q–Q plots (**Figure 6-7**) and goodness-of-fit tests and Normal distribution was checked only as a baseline comparison by Normal Q–Q plots (**Figure 8**). **Table 2** reveals the KS and chi-square p-values for the proposed models.

Table 2. Goodness-of-fit p-values for Gamma and Lognormal models

Breed	KS p (Gamma)	KS p (Lognormal)	Chi-square p (Gamma)	Chi-square p (Lognormal)
Mixed Breed	0.011	<0.001	0.228	<0.001
Shih Tzu	0.946	0.938	0.611	0.584
Staffordshire Bull Terrier	0.908	0.184	0.993	0.588

For the Mixed Breed, the histogram and Q–Q plots displayed several deviations in the upper tail, and the KS test resulted in a small p-value. These highlight that neither model fit the tail pattern precisely. The Lognormal model performed even worse by depicting clear curvature in its Q–Q plot and quite small goodness-of-fit p-figures.

Moving to Shih Tzu, both Gamma and Lognormal models presented uncertain results with the empirical distributions that followed the theoretical line closely even though large p-values was uncovered. Therefore, decisions for Shih Tzu stay less certain mostly owing to its limited observations.

Overall, the Gamma distribution appears to be the more logical choice, especially for the two larger datasets, whilst the Lognormal model is confirmed to be less appropriate.

2.5 Inference on Mean Rehoming Time

A t-interval was used for each breed since the population variance is unknown and must be derived from the data. Moreover, the t-distribution for both large and small

samples, but it is especially necessary for the small sample size of Shih Tzu breed, where the normal (z) approximation would not be suitable.

From a caterpillar plot (**Figure 9**), it is obvious that all three confidence intervals were centered around 19 weeks and lay entirely below the reference value of 27. In addition, the interval for Shih Tzu breed was noticeably wider mainly due to its small sample size but still far from 27. **Table 3** shows the results of the t-test performed.

Table 3. Confidence intervals for mean rehoming time (H_0: μ = 27 weeks)

Breed	Mean	95% CI (Lower)	95% CI (Upper)	Decision vs 27 weeks
Mixed Breed	19.11	18.41	19.82	Mean **< 27**, reject H_0
Shih Tzu	19.50	15.77	23.23	Mean **< 27**, reject H_0
Staffordshire Bull Terrier	19.46	18.01	20.91	Mean **< 27**, reject H_0

2.6 Comparison Between Breeds

In this part, pairwise comparisons of mean rehoming times were performed applying Welch's t-intervals because there are substantial differences in sample sizes, and their variances were not considered being equal. In all three comparisons, the confidence intervals for the difference in means have zero (**Table 4**) and this suggests that no statistically meaningful difference exists in average rehoming values between any pair of breeds. Although estimates for Shih Tzus are less precise, this imprecision does not suggest a significant difference and it is because of the small sample size.

Table 4. Pairwise comparisons of mean rehoming times (Welch CI for $\mu_1 - \mu_2$)

Comparison	Mean Difference	CI Lower	CI Upper	p-value
Mixed vs Shih Tzu	-0.3862	-4.1714	3.3990	0.8352
Mixed vs Staffordshire	-0.3496	-1.9608	1.2616	0.6697
Shih Tzu vs Staffordshire	0.0366	-3.9314	4.0046	0.9851

3. Discussion

Regarding limitations, the KS test for the 1st breed led to a small p-value mainly because of the large enough sample, improving the test sensitivity so that even minor discrepancies in the tail affects it. Besides, the warnings from the KS and chi-square tests occurred since rehoming values were recorded in whole weeks as integers. Thus, ties and low expected counts were formed in some histogram bins and these proposed issues decreased the reliability of the tests. For Shih Tzu, the limited sample size prevents acquiring clear conclusions, in spite of the models appearing to fit reasonably well according to the tests.

To conclude, the Gamma distribution can be the best fit for rehoming times, particularly in terms of two bigger samples (Staffordshire and Mixed Breed) and all breeds had average rehoming times of around 19 weeks, not 27 weeks. Statistical tests found no significant differences between breeds which might suggest practical importance, clarifying breed is not only the main factor to influence rehoming periods. For Future work, the usage of larger and more balanced data is required to examine other modelling approaches, and include additional predictors to capture rehoming variations in detail.

Acknowledgements

I would like to thank the module teaching team for providing guidance and materials on the statistical methods used in this coursework.

Appendices

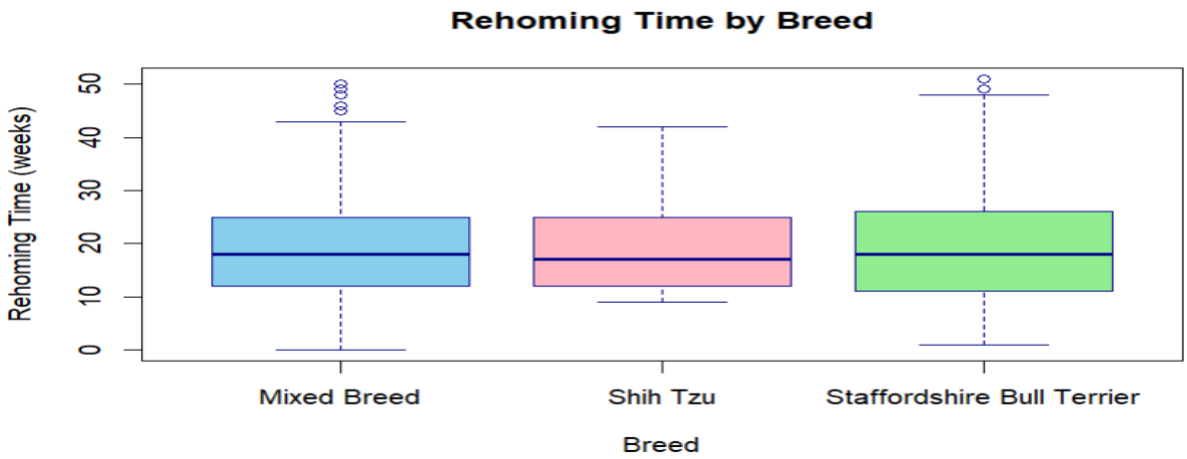

Figure 1. Boxplot of rehoming time by breed, showing the distribution, median, and presence of upper-tail outliers for each group.

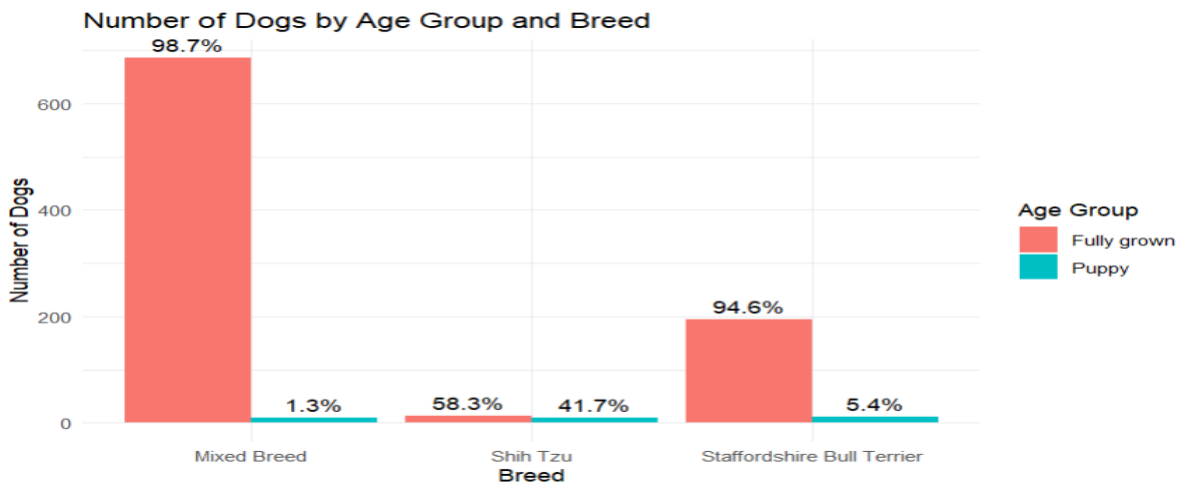

Figure 2. Bar chart showing age-group distribution across breeds, with relative percentages displayed.

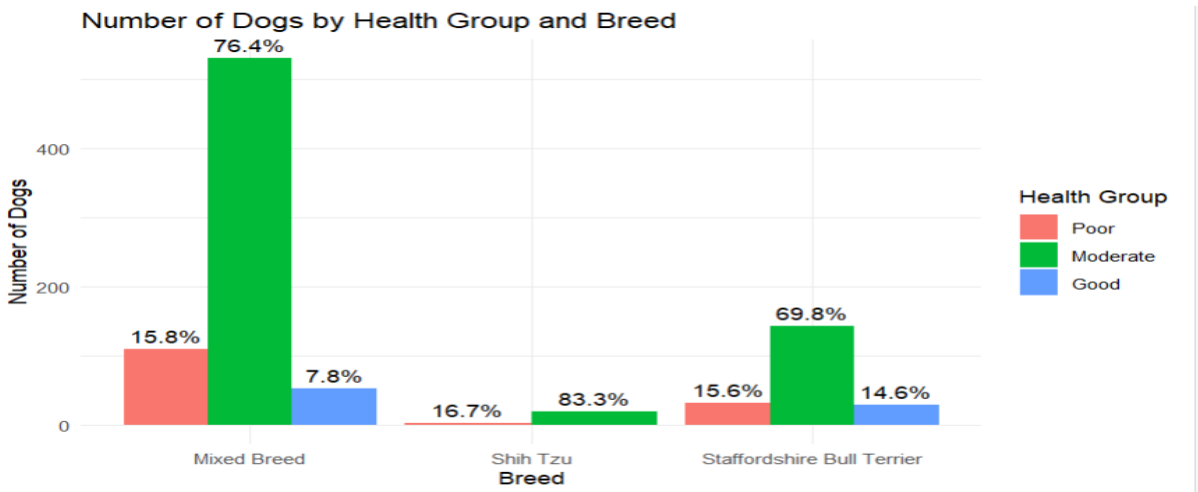

Figure 3. Bar chart showing the distribution of dogs across three health categories (Poor, Moderate, Good) within each breed.

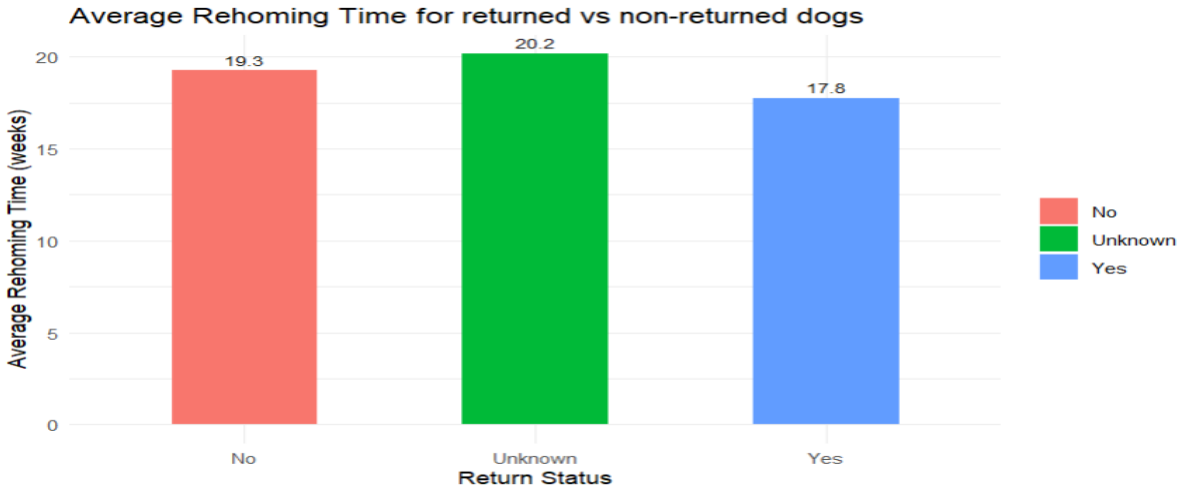

Figure 4. Bar chart showing the average rehoming time for dogs grouped by return status ("No," "Unknown," and "Yes").

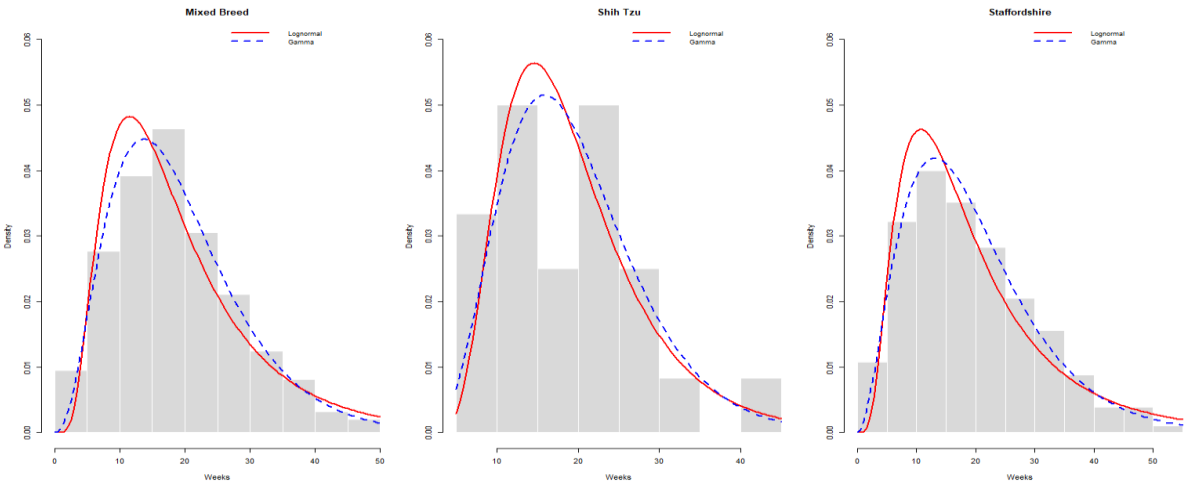

Figure 5. Histograms of rehoming time for each breed with fitted Lognormal and Gamma density curves overlaid for comparison.

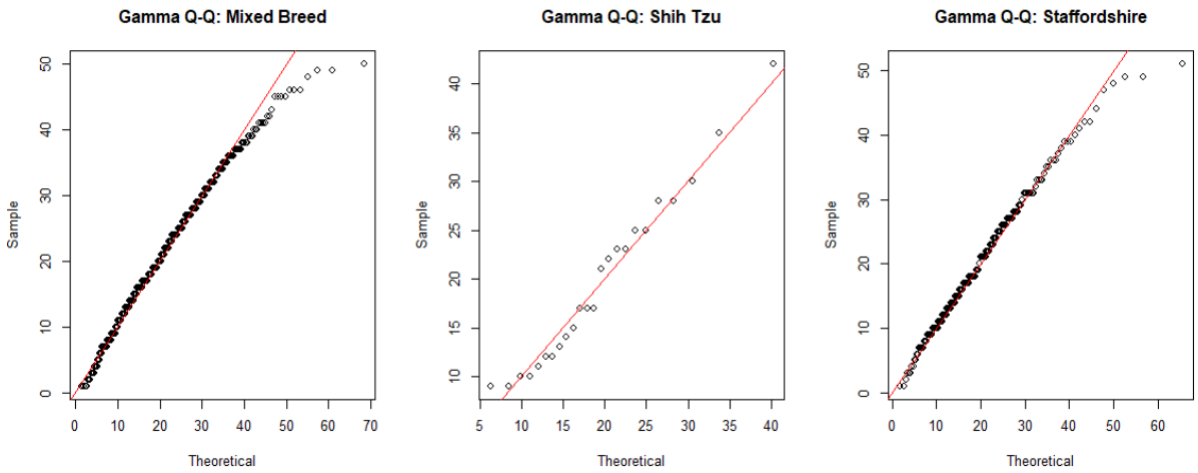

Figure 6. Gamma Q–Q plots for each breed comparing sample quantiles with theoretical Gamma quantiles.

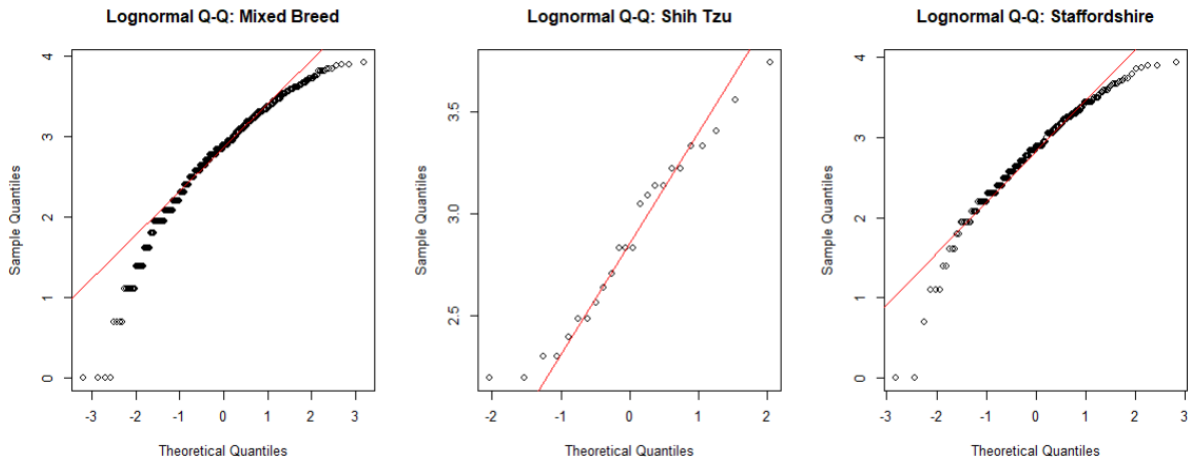

Figure 7. Lognormal Q–Q plots for each breed assessing the adequacy of the Lognormal model.

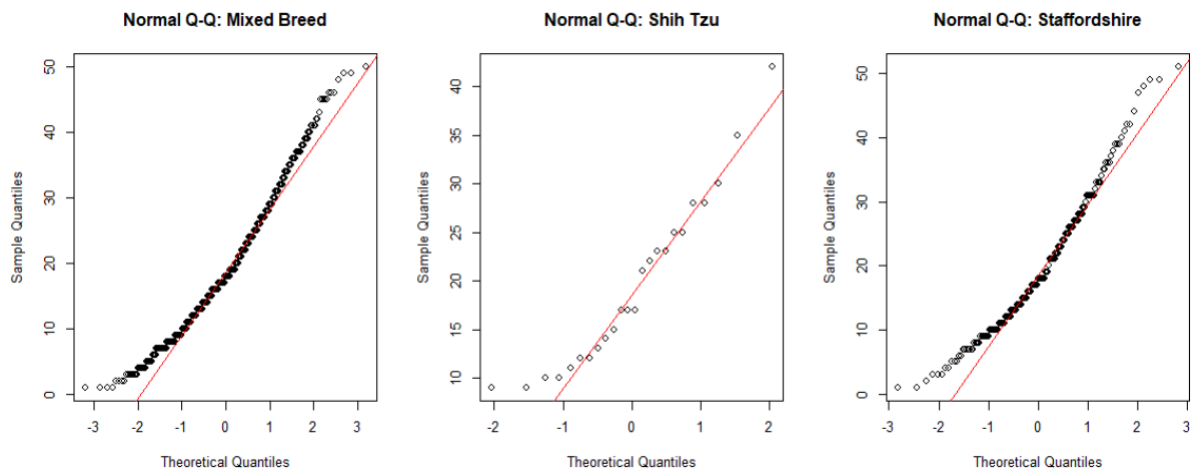

Figure 8. Normal Q–Q plots for each breed assessing departure from Normality in rehoming times.

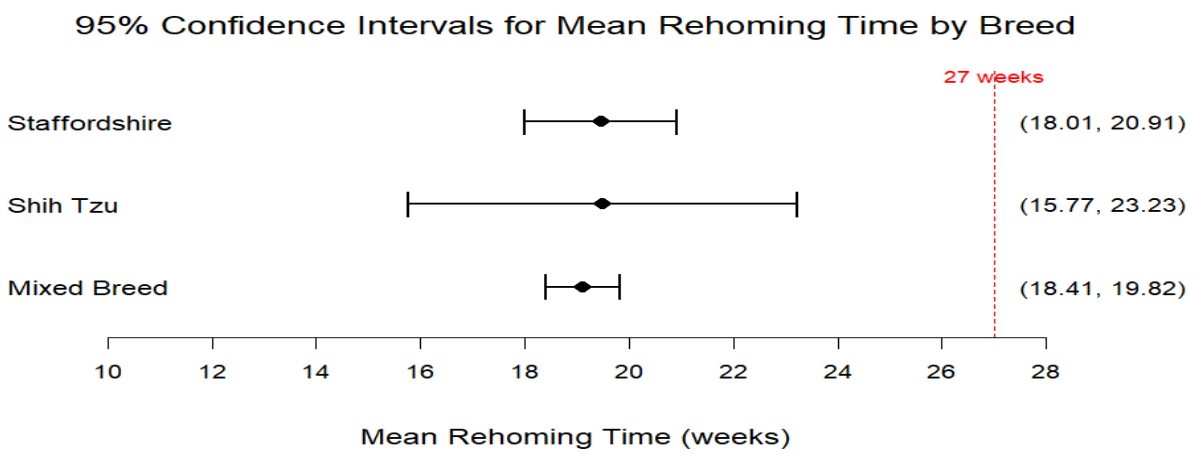

Figure 9. Caterpillar plot showing the 95% confidence intervals for the mean rehoming time of each breed. The dashed vertical line at 27 weeks represents the reference value used for hypothesis testing.

Codes

```
#-----------------------------------------------------------------------------

# Used Libraries


if (!require(e1071)) install.packages("e1071")

if (!require(MASS)) install.packages("MASS")

if (!require(ggplot2)) install.packages("ggplot2")

if (!require(dplyr)) install.packages("dplyr")

if (!require(knitr)) install.packages("knitr")


library(e1071)

library(MASS)

library(ggplot2)

library(dplyr)

library(knitr)


#-----------------------------------------------------------------------------

# 1. LOADING DATA AND QUICK INSPECTION


createsample(202007327)

save(mysample, file = "mysample.RData")


head(mysample)

str(mysample)

summary(mysample)

table(mysample$Breed)


#-----------------------------------------------------------------------------

# DATA CLEANING


df <- mysample
```

```
total_rows <- nrow(df)
total_rows

# Removing duplicates
df$Rehomed[df$Rehomed == 99999] <- NA
duplicates_removed <- sum(duplicated(df))
df <- df[!duplicated(df), ]

# Remove missing Rehomed times
missing_rehomed <- sum(is.na(df$Rehomed))
df <- df[!is.na(df$Rehomed), ]

# Removing missing Breed
missing_breed <- sum(is.na(df$Breed))
df <- df[!is.na(df$Breed), ]

# Removing negative time values
negative_count <- sum(df$Rehomed < 0 | df$Visited < 0)
df <- df[df$Rehomed >= 0 & df$Visited >= 0, ]

df_clean <- df
cleaned_rows <- nrow(df_clean)
total_removed <- total_rows - cleaned_rows

cat("Total removed:", total_removed, "\n")

#-----------------------------------------------------------------------------
# SPLITING DATA BY BREED

mixed    <- df_clean[df_clean$Breed == "Mixed Breed", ]
shihtzu  <- df_clean[df_clean$Breed == "Shih Tzu", ]
stafford <- df_clean[df_clean$Breed == "Staffordshire Bull Terrier", ]
```

```
x1 <- mixed$Rehomed[mixed$Rehomed > 0]

x2 <- shihtzu$Rehomed[shihtzu$Rehomed > 0]

x3 <- stafford$Rehomed[stafford$Rehomed > 0]


#-------------------------------------------------------------------------------

# Summary tables


summary_table <- data.frame(

  Breed = c("Mixed Breed", "Shih Tzu", "Staffordshire Bull Terrier"),

  n     = c(length(x1), length(x2), length(x3)),

  Mean  = c(mean(x1), mean(x2), mean(x3)),

  SD    = c(sd(x1), sd(x2), sd(x3)),

  Skewness = c(skewness(x1), skewness(x2), skewness(x3))

)


summary_table

#-------------------------------------------------------------------------------

# GRAPHICAL SUMMARIES


# Boxplot for each breed

boxplot(Rehomed ~ Breed, data = df_clean,

        main = "Rehoming Time by Breed",

        xlab = "Breed", ylab = "Weeks",

        col = c("skyblue","lightpink","lightgreen"))


# Histograms for each breed

hist(mixed$Rehomed, main="Mixed Breed Rehoming Time",

     xlab="Weeks", col="lightgrey", border="white")


hist(shihtzu$Rehomed, main="Shih Tzu Rehoming Time",

     xlab="Weeks", col="lightgrey", border="white")
```

```
hist(stafford$Rehomed, main="Staffordshire Bull Terrier Rehoming Time",
    xlab="Weeks", col="lightgrey", border="white")

# Age distribution  graph
df_age <- df_clean %>%
  count(Breed, Age) %>%
  group_by(Breed) %>%
  mutate(Percentage = n/sum(n)*100)

ggplot(df_age, aes(x=Breed, y=n, fill=Age)) +
  geom_bar(stat="identity", position=position_dodge()) +
  geom_text(aes(label=paste0(round(Percentage,1),"%")),
          position=position_dodge(0.9), vjust=-0.5) +
  labs(title="Age Group Distribution by Breed", x="Breed", y="Count") +
  theme_minimal()

# Health conditions graph
df_clean$Health_Group <- cut(df_clean$Health,
                        breaks=c(0,40,70,100),
                        labels=c("Poor (<40)","Moderate (<70)","Good (>=70)"),
                        include.lowest=TRUE)

df_health <- df_clean %>%
  count(Breed, Health_Group) %>%
  group_by(Breed) %>%
  mutate(Percentage = n/sum(n)*100)

ggplot(df_health, aes(x=Breed, y=n, fill=Health_Group)) +
  geom_bar(stat="identity", position=position_dodge()) +
  geom_text(aes(label=paste0(round(Percentage,1),"%")),
          position=position_dodge(0.9), vjust=-0.5) +
```

```
  labs(title="Health Group Distribution by Breed",
      x="Breed", y="Count", fill="Health Group") +
  theme_minimal()

# Return status graph
ggplot(df_clean %>% mutate(Returned_clean = ifelse(is.na(Returned),"Unknown",Returned)),
      aes(x=Returned_clean, y=Rehomed, fill=Returned_clean)) +
  stat_summary(fun="mean", geom="bar", width=0.5) +
  stat_summary(fun="mean", geom="text",
            aes(label=round(..y..,1)), vjust=-0.5) +
  labs(title="Average Rehoming Time by Return Status",
      x="Return Status", y="Weeks") +
  theme_minimal() + theme(legend.position="none")

#-----------------------------------------------------------------------------
# FITING MODELS

fit_lognorm_mixed    <- fitdistr(x1, "lognormal")
fit_gamma_mixed      <- fitdistr(x1, "gamma")

fit_lognorm_shihtzu  <- fitdistr(x2, "lognormal")
fit_gamma_shihtzu    <- fitdistr(x2, "gamma")

fit_lognorm_stafford <- fitdistr(x3, "lognormal")
fit_gamma_stafford   <- fitdistr(x3, "gamma")

# -----------------------------------------------------------------------------
# MODEL CHECKING (HISTOGRAMS + FITTED CURVES + QQ PLOTS)

plot_fitted <- function(data, fit_lnorm, fit_gamma, title){
  hist(data, probability=TRUE, main=title, xlab="Weeks",
      col="grey85", border="white", ylim=c(0, 0.06))
```

```
  # Lognormal curve
  curve(dlnorm(x, fit_lnorm$estimate[1], fit_lnorm$estimate[2]),
      add=TRUE, col="red", lwd=2)

  # Gamma curve
  curve(dgamma(x, fit_gamma$estimate[1], fit_gamma$estimate[2]),
      add=TRUE, col="blue", lwd=2, lty=2)

  legend("topright",
       legend = c("Lognormal", "Gamma"),
       col    = c("red", "blue"),
       lwd    = 2,
       lty    = c(1,2),
       bty    = "n",
       cex    = 0.9)
}

par(mfrow = c(1, 3), mar=c(4,4,3,1), oma=c(0,0,2,0))
plot_fitted(x1, fit_lognorm_mixed, fit_gamma_mixed, "Mixed Breed")
plot_fitted(x2, fit_lognorm_shihtzu, fit_gamma_shihtzu, "Shih Tzu")
plot_fitted(x3, fit_lognorm_stafford, fit_gamma_stafford, "Staffordshire")
par(mfrow = c(1,1))

# Normal Q-Q plots
par(mfrow=c(1,3))
qqnorm(x1, main="Normal Q-Q: Mixed Breed"); qqline(x1, col="red")
qqnorm(x2, main="Normal Q-Q: Shih Tzu"); qqline(x2, col="red")
qqnorm(x3, main="Normal Q-Q: Staffordshire"); qqline(x3, col="red")
par(mfrow=c(1,1))

# Gamma Q-Q plots
```

```
par(mfrow=c(1,3))
qqplot(qgamma(ppoints(length(x1)),
        shape=fit_gamma_mixed$estimate["shape"],
        rate =fit_gamma_mixed$estimate["rate"]),
    x1, main="Gamma Q-Q: Mixed Breed",
    xlab="Theoretical", ylab="Sample"); abline(0,1,col="red")

qqplot(qgamma(ppoints(length(x2)),
        shape=fit_gamma_shihtzu$estimate["shape"],
        rate =fit_gamma_shihtzu$estimate["rate"]),
    x2, main="Gamma Q-Q: Shih Tzu",
    xlab="Theoretical", ylab="Sample"); abline(0,1,col="red")

qqplot(qgamma(ppoints(length(x3)),
        shape=fit_gamma_stafford$estimate["shape"],
        rate =fit_gamma_stafford$estimate["rate"]),
    x3, main="Gamma Q-Q: Staffordshire",
    xlab="Theoretical", ylab="Sample"); abline(0,1,col="red")
par(mfrow=c(1,1))

# Lognormal Q-Q plots
par(mfrow=c(1,3))
qqnorm(log(x1), main="Lognormal Q-Q: Mixed Breed"); qqline(log(x1), col="red")
qqnorm(log(x2), main="Lognormal Q-Q: Shih Tzu"); qqline(log(x2), col="red")
qqnorm(log(x3), main="Lognormal Q-Q: Staffordshire"); qqline(log(x3), col="red")
par(mfrow=c(1,1))

#-----------------------------------------------------------------------------
# Model Testing

# KS tests
ks1_gamma   <- ks.test(x1, "pgamma", shape=fit_gamma_mixed$estimate[1], rate=fit_gamma_mixed$estimate[2])
```

```
ks1_lognorm <- ks.test(x1, "plnorm", meanlog=fit_lognorm_mixed$estimate[1],
sdlog=fit_lognorm_mixed$estimate[2])


ks2_gamma   <- ks.test(x2, "pgamma", shape=fit_gamma_shihtzu$estimate[1], rate=fit_gamma_shihtzu$estimate[2])

ks2_lognorm <- ks.test(x2, "plnorm", meanlog=fit_lognorm_shihtzu$estimate[1],
sdlog=fit_lognorm_shihtzu$estimate[2])


ks3_gamma   <- ks.test(x3, "pgamma", shape=fit_gamma_stafford$estimate[1], rate=fit_gamma_stafford$estimate[2])

ks3_lognorm <- ks.test(x3, "plnorm", meanlog=fit_lognorm_stafford$estimate[1],
sdlog=fit_lognorm_stafford$estimate[2])


# Chi-square tests

chisq_test <- function(x, fit_gamma, fit_lnorm){

  h <- hist(x, breaks=10, plot=FALSE)


  p_gamma <- pgamma(h$breaks[-1], shape=fit_gamma$estimate[1], rate=fit_gamma$estimate[2]) -

    pgamma(h$breaks[-length(h$breaks)], shape=fit_gamma$estimate[1], rate=fit_gamma$estimate[2])

  p_gamma <- p_gamma / sum(p_gamma)


  p_lognorm <- plnorm(h$breaks[-1], meanlog=fit_lnorm$estimate[1], sdlog=fit_lnorm$estimate[2]) -

    plnorm(h$breaks[-length(h$breaks)], meanlog=fit_lnorm$estimate[1], sdlog=fit_lnorm$estimate[2])

  p_lognorm <- p_lognorm / sum(p_lognorm)


  list(

    gamma = chisq.test(h$counts, p=p_gamma),

    lognorm = chisq.test(h$counts, p=p_lognorm)

  )

}


chisq1 <- chisq_test(x1, fit_gamma_mixed, fit_lognorm_mixed)

chisq2 <- chisq_test(x2, fit_gamma_shihtzu, fit_lognorm_shihtzu)

chisq3 <- chisq_test(x3, fit_gamma_stafford, fit_lognorm_stafford)
```

```
#-----------------------------------------------------------------------------
# CONFIDENCE INTERVALS

t1 <- t.test(x1, mu=27)
t2 <- t.test(x2, mu=27)
t3 <- t.test(x3, mu=27)
t1
t2
t3

#-----------------------------------------------------------------------------
# CI Plot for Dog Rehoming Times

breeds <- c("Mixed Breed", "Shih Tzu", "Staffordshire")
means  <- c(19.11, 19.50, 19.46)
lower  <- c(18.41, 15.77, 18.01)
upper  <- c(19.82, 23.23, 20.91)

verticalpos <- 1:3
cex <- 1.2
par(oma=c(0,0,0,0), mar=c(6,2,4,4))
plot(x=1, y=0, type="n", xaxt='n', yaxt='n',
    xlab="", ylab="", bty="n",
    xlim=c(10, 28), ylim=c(0.5, 3.5))
mtext("95% Confidence Intervals for Mean Rehoming Time by Breed",
     side = 3, line = 1.5, cex = 1.4, font = 1)

# line for 27 weeks
abline(v = 27, col="red", lty=2, lwd=1.5)
text(27, 3.35, "27 weeks", col="red", cex=0.9, pos=3)
points(means, verticalpos, pch=16, cex=1.3)
```

```
for (i in 1:3) {
  lines(c(lower[i], upper[i]), c(verticalpos[i], verticalpos[i]), lwd=2)
  lines(c(lower[i], lower[i]), c(verticalpos[i] - 0.15, verticalpos[i] + 0.15), lwd=2)
  lines(c(upper[i], upper[i]), c(verticalpos[i] - 0.15, verticalpos[i] + 0.15), lwd=2)
}

axis(side=1, at=seq(10, 28, 2), cex.axis=1)
mtext("Mean Rehoming Time (weeks)", side=1, line=4, cex=1.2)
mtext(text = breeds, side=2, line=2, at=verticalpos, las=1, cex=1.1, adj=0)
ci_labels <- paste0("(", lower, ", ", upper, ")")
mtext(text = ci_labels, side=4, line=-2, at=verticalpos, las=1, cex=1)

#-------------------------------------------------------------------------------
# BREED COMPARISONS

tt12 <- t.test(x1, x2, var.equal=FALSE)
tt13 <- t.test(x1, x3, var.equal=FALSE)
tt23 <- t.test(x2, x3, var.equal=FALSE)
tt12
tt13
tt23
```

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School of Mathematics

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I declare that the attached submission is my own work.

Where the work of others has contributed to my work, I have given full acknowledgement using the appropriate referencing conventions for my programme of study.

I confirm that the attached submission has not been submitted for marks or credits in a different module or for a different qualification or completed prior to entry to the University.

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Student Signature: **Student Number:** 202007327

Student Name: Ali Sharifzade **Date:** 12 December 2025

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